



Advanced TruAge Report

Christopher Hughett

Age: 37 | Sex: Male

ID#: N49KZCE4ZM494952

Collected: 05/11/2026 | Reported: 05/31/2026

Reports corresponding to samples processed after November 10, 2025 have been generated using our updated analysis platform and newly trained algorithms. These improvements may result in minor differences compared to prior reports, reflecting advances in our technology rather than biological changes.

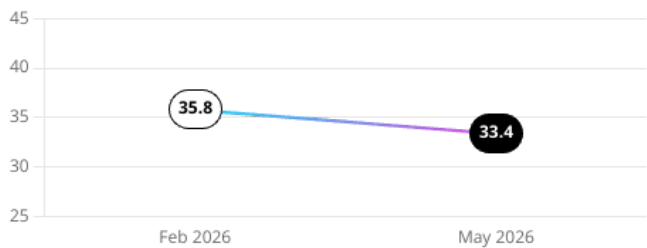
CLIA Lab Director: Melissa Keinath, PhD FACMG

TruDiagnostic

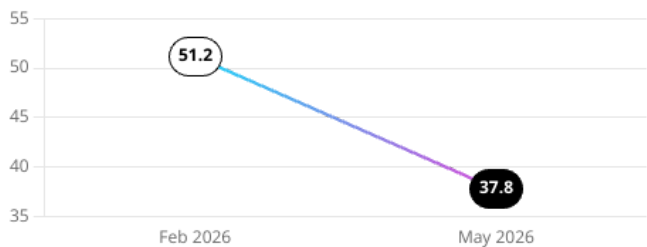
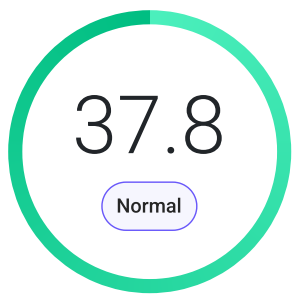
881 Corporate Drive

Lexington, KY 40503

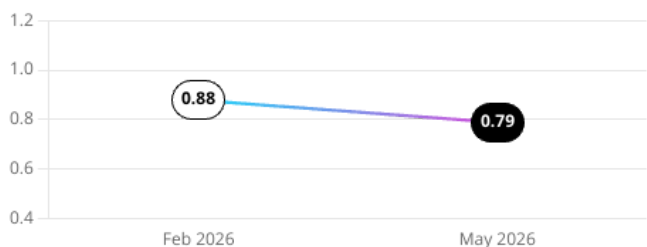
OMICm Age



Symphony Age



Pace of Aging



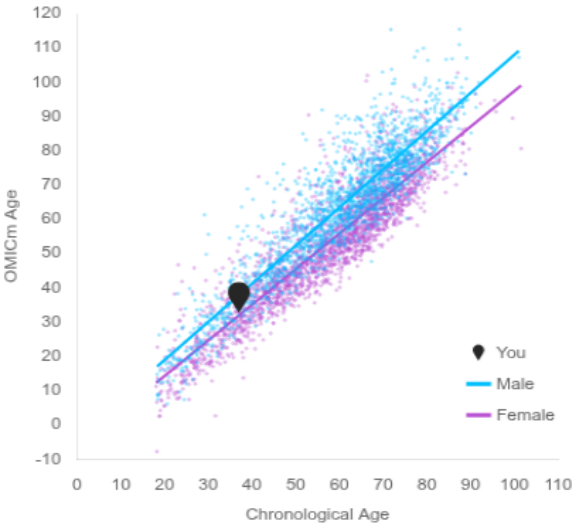
OMICm Age

33.4

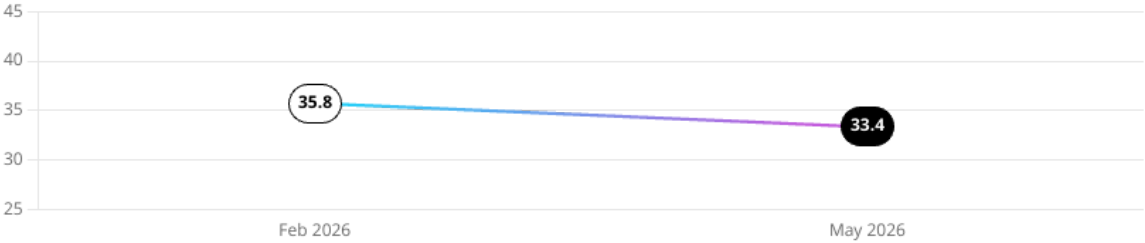
Younger ↘

29.4%

Your OMICm Age is **lower** than 70.6% of people of the same age and sex.



Results Over Time



DISCLAIMER: The population graph and percentile for OMICm Age are based on observed and validated data patterns from an equal distribution of Harvard research participants and TruDiagnostic clients to emulate a population of average health.

Most Actionable Epigenetic Biomarkers

Epigenetic Biomarker (EBs) are compared against a balanced reference dataset composed of individuals from both the Harvard and TruDiagnostic cohorts. We identify individuals of the same sex and a similar age (within ± 5 years) to create a personalized comparison group. Above the 80th percentile: Indicates your outcome is higher than 80% of individuals in the cohort. Below the 20th percentile: Indicates your outcome is lower than 80% of individuals in the cohort.

All of your EBs are in the correct range.

Find the list of your Epigenetic Biomarkers at the end of the report.

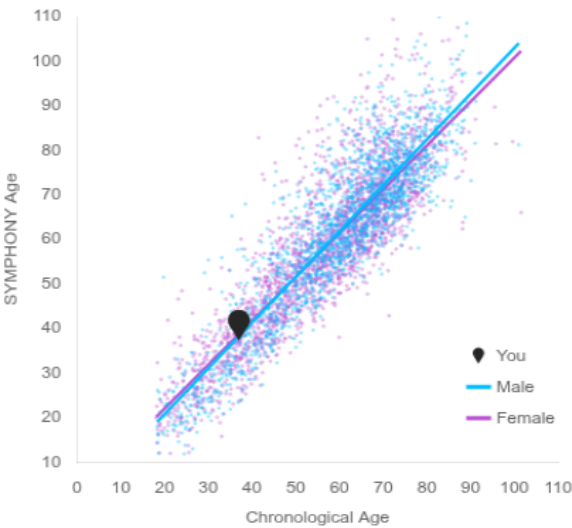
This advanced approach dives into the age of **11 distinct organ systems**.

SYMPHONY Age

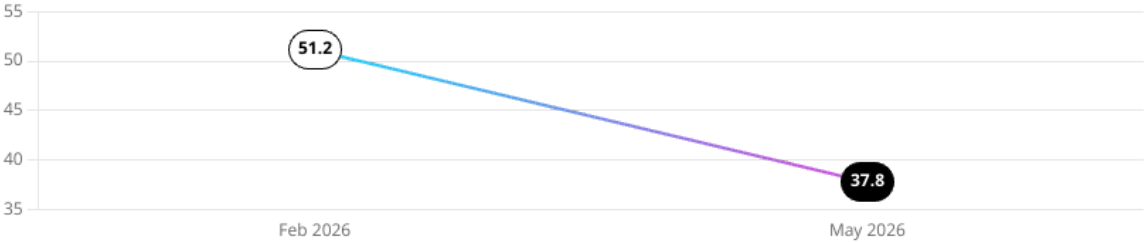
37.8

65.5%

Your SYMPHONY Age is **lower** than 34.5% of people of the same age and sex.



Results Over Time



NOTE: The percentile for SYMPHONY Age is based on observed and validated data patterns from an equal distribution of Harvard research participants and TruDiagnostic clients to emulate a population of average health. Our SYMPHONYAge algorithm was re-trained in November 2025 to enhance accuracy and performance. Samples processed before November 10, 2025, were reprocessed in December 2025, to reflect this updated model.

Compare the age of **11 distinct organ systems** versus your chronological age

Below Chronological Age Chronological Age Above Chronological Age



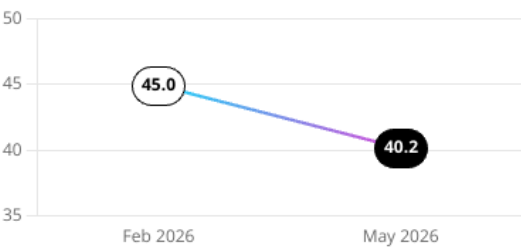
How was this clock created and designed?

Many clocks have used blood based biomarkers and their changes as we get older to predict OMICm age. For SYMPHONY Age, the researchers did this with biomarkers which were specific for each organ system. While previous clocks (like PhenoAge) did this with 9 blood markers, SYMPHONY Age incorporates 133 biomarkers for training to develop this clock. In addition to having a large and detailed number of clinical datapoints, this study is also one of the largest for clock development with approximately 8,000 participants. Together, this gives us an incredibly robust clock associated with organ specific clinical biomarkers which is highly predictive of aging outcomes.

Systems Related Biomarkers

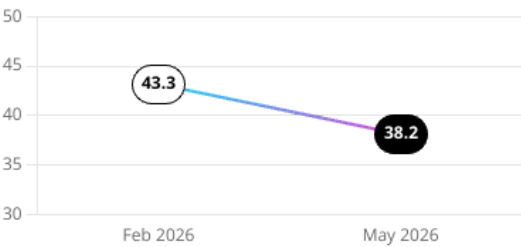
Blood

Biomarkers: Ferritin, Hematocrit, Hemoglobin, Mean Corpuscular Hemoglobin, Mean Corpuscular Hemoglobin Conc, Mean Corpuscular Volume, Mean Platelet Volume, Platelet Distribution width, Platelet Count, Red Blood Cell Count, Red Cell Distribution Width



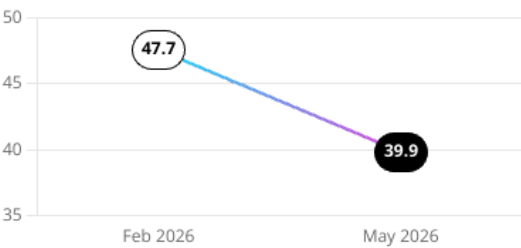
Brain

Biomarkers: Homocysteine, BDNF (serum), Clusterin, Stroke, Total mental status summary score, Total cognition summary score, Immediate word recall, Delayed word recall, Total word recall summary score, Serial 7s



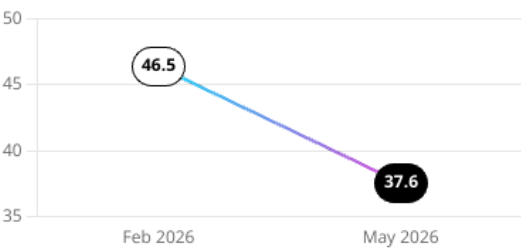
Inflammation

Biomarkers: Ferritin, C-Reactive Protein, Transforming Growth Factor Beta, Interleukin 10, Interleukin 1 Receptor Antagonist, Interleukin 6, Tumor Necrosis Factor Receptor 1



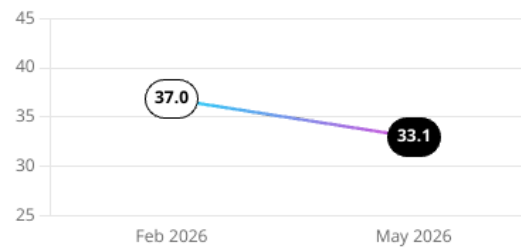
Heart

Biomarkers: Shortness of breath while awake, PCcomponents of Grimage, Previous High Blood Pressure, Previous Heart Attack, Previous Stroke, Homocysteine, BMI



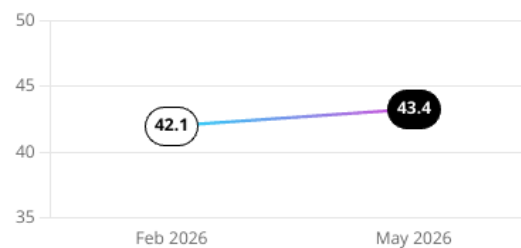
Hormone

Biomarkers: IGF-1, DHEAS



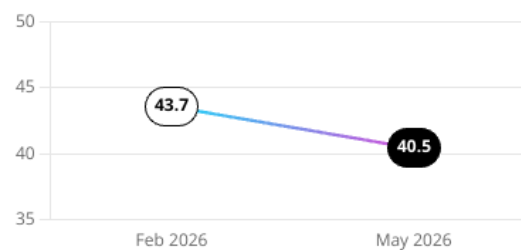
Immune

Biomarkers: Eosinophil Count, Lymphocyte Count, Monocyte Count, Neutrophil Count, Percent Basophils, Percent Eosinophils, Percent Lymphocytes, Percent Monocytes, White Blood Cell Count, Myeloid Dendritic cells (DC-M) Percentage, Plasmacytoid, Dendritic Cells (DC-P) Percentage, NK Cells: CD56HI Percentage, NK Cells: CD56LO Percentage, CD16- Monocytes Percentage, CD16+ Monocytes Percentage, B Cells Percentage, CD8+ T Cells: Central Memory (CM) Percentage, CD4+ T Cells: Central Memory (CM) Percentage, CD8+ T Cells Percentage, CD8+ T Cells: (TemRA) Percentage, CD4+ T Cells: (TemRA) Percentage, CD4+ T Cells Percentage, IgD+ Memory B Cells Percentage, IgD- Memory B Cells Percentage, CD8+ T Cells: Naïve Percentage, CD4+ T Cells: Naïve Percentage, T Cells Percentage, Naïve B Cells Percentage, CD8+T Cells: Effector Memory (Tem) Percentage, CD4+ T Cells: Effector Memory (Tem) Percentage, Natural Killer (NK) Cells Percentage, Monocytes Percentage, Dendritic Cells Percentage



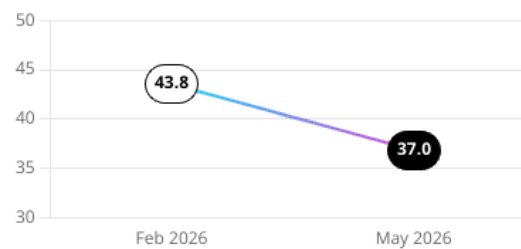
Kidney

Biomarkers: Albumin, Urea Nitrogen, Chloride, Bicarbonate, Creatinine, Cystatin C, Potassium, Sodium



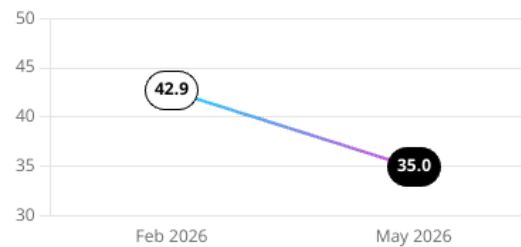
Liver

Biomarkers: Albumin, Alkaline Phosphatase, ALT, AST, Bilirubin, Total Protein



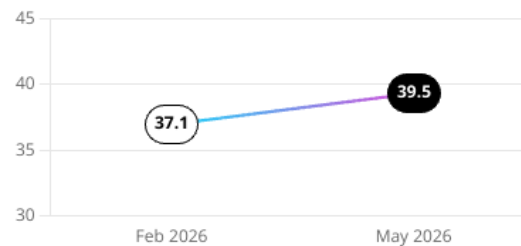
Metabolic

Biomarkers: PCSmoking-packyears, Previous Diabetes, C-Reactive Protein, Glucose-Fasting, HDL-Cholesterol, LDL-Cholesterol, Triglycerides, Interleukin-6, BMI, Waist circumference



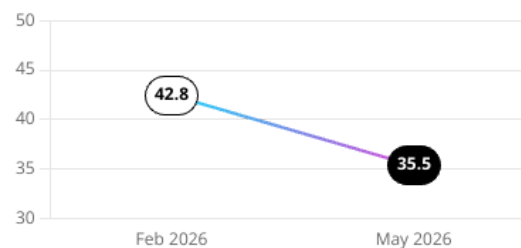
Lung

Biomarkers: Peak expiratory flow, bicarbonate, chronic lung disease, shortness of breath while awake, persistent wheezing, cough, or bringing up phlegm, PCSmoking-packyears

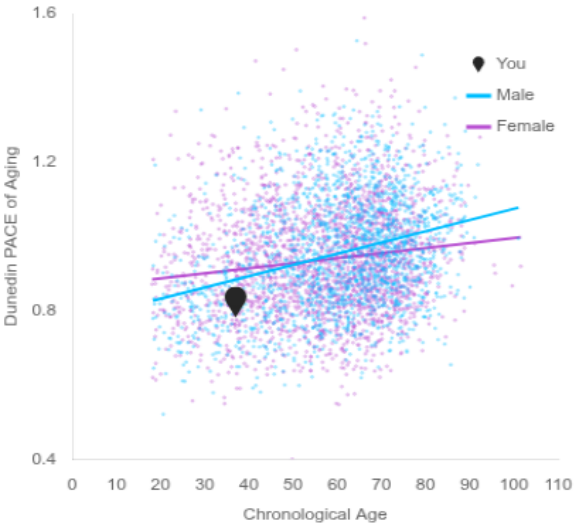
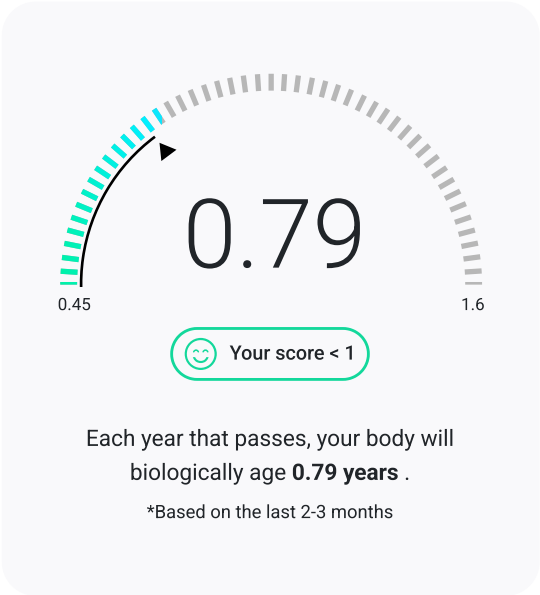


MusculoSkeletal

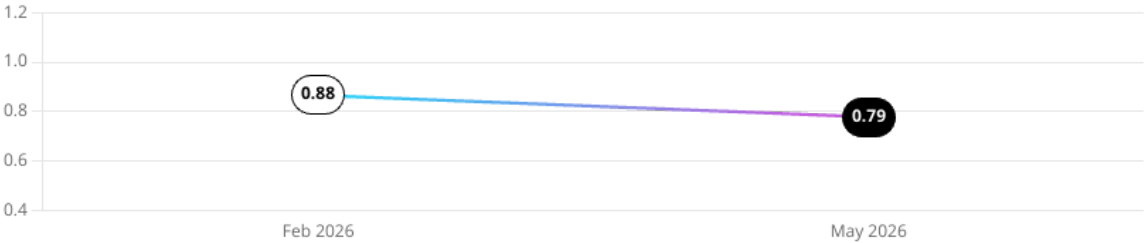
Biomarkers: Vitamin D3, Dehydroepiandrosterone sulphate, IGF-1, Arthritis, Height, Weight, BMI, some diff-mobility, hand grip strength maximum measurement, semi tandem balance test time, timed walk test time, hand grip strength-left hand hand grip strength-right hand, had back problems, some diff-stoop/kneel/crouch, diff-stoop/kneel/crouch, diff-walk one block, diff-walk sev blocks, some diff-walk one block, some diff-walk sev blocks, diff-climb sev flt stair, diff-climb one flt stair, some diff-clmb sev flt str, some diff-clmb 1 flt stair, diff-get up fr chair, some diff-get up fr chair, diff-reach/extnd arms up, some diff-rch/xtnnd arms up, diff-lift/carry 10lbs, some diff-lift/carry 10lbs, side-by-side balance test time, full tandem balance test time, Sum of 7 different functional tests, Combination of all balance scores



Your Rate of Aging will change based on your lifestyle interventions. Be sure to retest every 3 months to track your progress.



Results Over Time



DISCLAIMER: The population graph for DunedinPACE of Aging is based on observed and validated data patterns from an equal distribution of Harvard research participants and TruDiagnostic clients to emulate a population of average health.

Immune Health

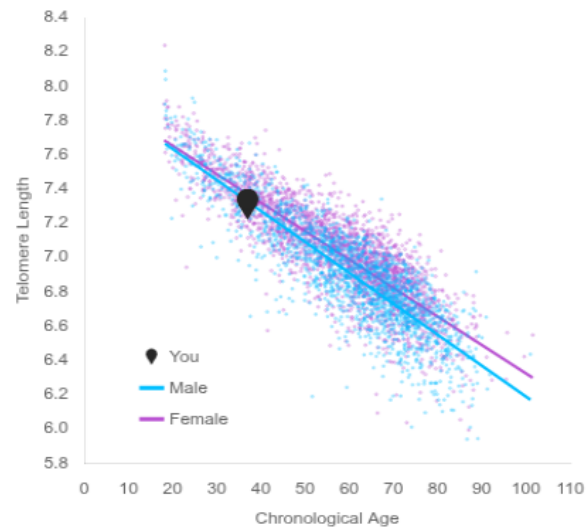
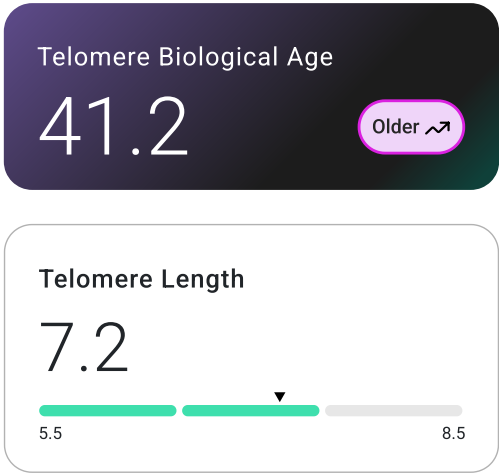


CD4/CD8T cell ratio is incredibly informative on disease. The normal range (ratio between 1.0 and 4.0, corresponding to the 30th–80th percentiles) helps to contextualize where your value falls. A low or inverted CD4/CD8 ratio is an immune-risk phenotype and is associated with altered immune function, immune senescence, and chronic inflammation.

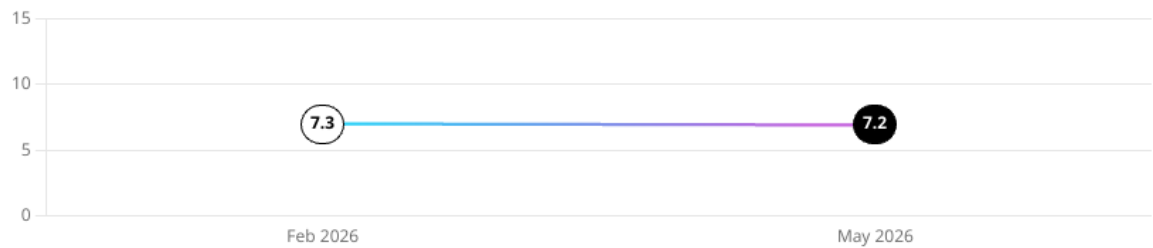


The Neutrophil-to-Lymphocyte Ratio (NLR) is obtained by dividing the number of neutrophils by the number of lymphocytes. During physiological stress, neutrophil count increases while lymphocyte count decreases. Physiological stress, driven by illness, inflammation, or psychological stress, can elevate NLR. Therefore, NLR elevation is not exclusive to infection or inflammation but can result from any form of physiological stress, including everyday stress and poor recovery or stress management.

Shorter telomere length and low telomerase activity are correlated with several chronic preventable diseases.



Results Over Time



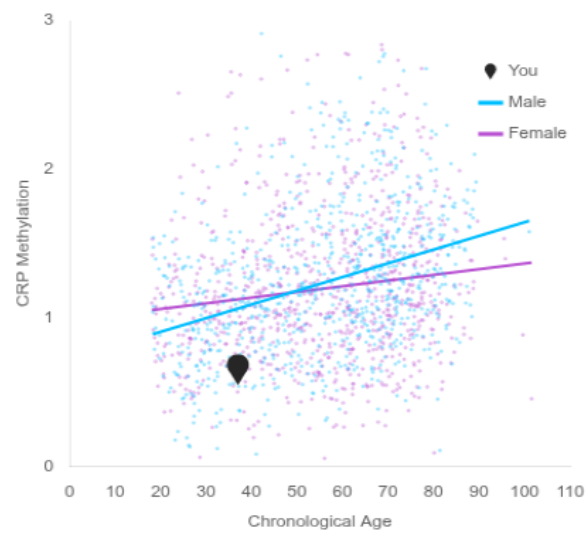
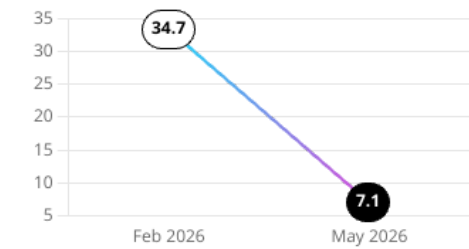
DISCLAIMER: Telomere length is mostly determined based on predetermined genetic function. The population graph is based on observed and validated data patterns from an equal distribution of Harvard research participants and TruDiagnostic clients to emulate a population of average health.

Inflammation



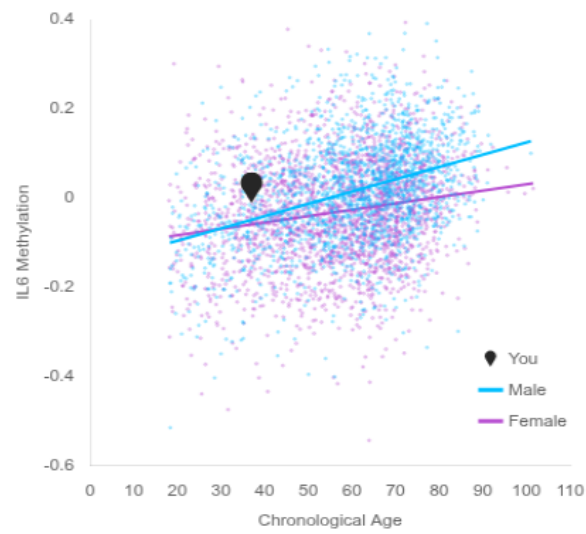
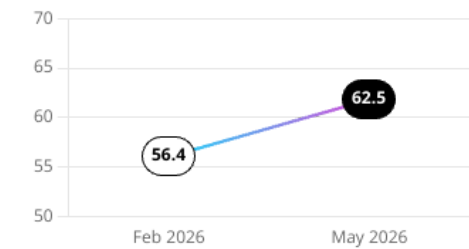
Your CRP methylation level is lower than 92.9% of the population at your same age and sex.

CRP is produced by the liver in response to acute inflammation. DNAm CRP has an inverse relationship with cognitive functions such as memory, speed, and visuospatial functions.



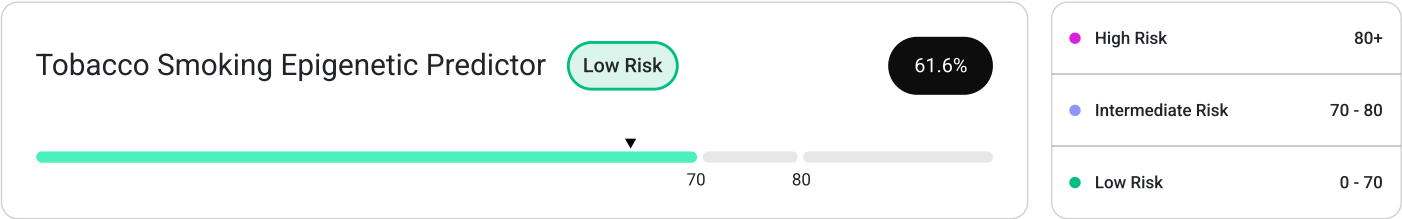
Your IL-6 methylation level is lower than 37.5% of the population at your same age and sex.

IL-6 is a widely used marker of inflammation and circulating levels of the cytokine typically rise in older age. DNAm IL-6 is positively associated with BMI, self-reported smoking status, and alcohol intake.

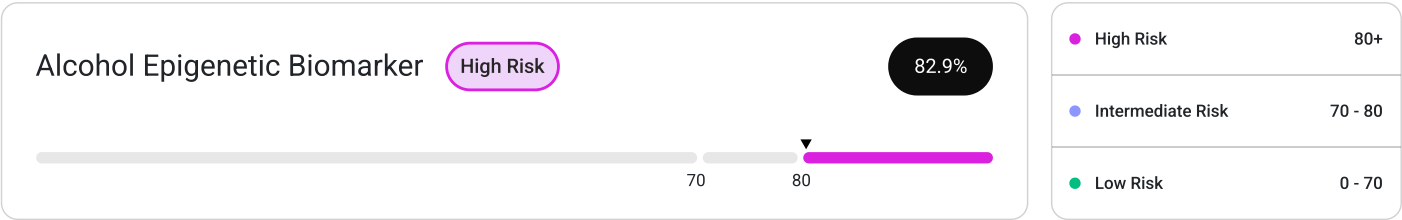


DISCLAIMER: The population graph and percentile are based on observed and validated data patterns from an equal distribution of Harvard research participants and TruDiagnostic clients to emulate a population of average health.

Relative Smoking Risk



Relative Alcohol Risk



Additional Epigenetic Biomarkers

Red Blood Cell Distribution Width

Normal

▼

080

5.9%

Function

Red Blood Cell Width (RDW): Measures the variation in red blood cell size.

Recommendations

- Correct nutrient deficiencies, especially B12, folate, iron, and vitamin C
- Manage chronic illness and inflammation
- Consider omega-3s
- Avoid alcohol and smoking

⊗ Vitamin B12

⊗ Folate

⊗ Iron

⊗ Vitamin C

⊗ Omega-3 Fatty Acids

✓ Red meat

✓ Beans

✓ Fortified cereals

✓ Dairy

✓ Meat

✓ Peppers

✓ Leafy green vegetables

✓ Citrus fruits

✓ Tomatoes

Albumin

Normal

20100

▼

83.5%

Function

Albumin: A protein in the blood that helps maintain oncotic pressure and transport substances.

Recommendations

- If applicable, consume high-protein foods
- Diversify protein sources
- Consider vitamin D and omega-3-rich foods or supplements.
- Manage liver and kidney health
- Limit nephrotoxic medications

⊗ Vitamin D

⊗ Omega-3 Fatty Acids

✓ Lean meats

✓ Fish

✓ Poultry

✓ Eggs

✓ Dairy

✓ Legumes

✓ Nuts

Fasting Glucose

Normal

▼

080

3.1%

Function

Glucose: Blood sugar level; essential for assessing metabolic health and diagnosing diabetes.

Recommendations

- Control carbohydrate intake when applicable
- Stay active
- Manage weight
- Avoid sugar
- Manage stress
- Consider medications if advised by your physician
- Consider supplements like ALA, berberine, or dihydroberberine

⊗ Alpha Lipoic Acid

⊗ Berberine

⊗ Dihydroberberine

✓ Lean proteins

✓ Healthy fats

✓ Fiber-rich foods

Hemoglobin

Normal

20100

▼

92.2%

Function

Hemoglobin: A protein in red blood cells that carries oxygen; critical for evaluating oxygen transport and anemia.

Recommendations

- Correct nutrient deficiencies, especially folate, iron, B12, and vitamin C.
- Consider EPO in the setting of CKD
- Manage chronic conditions
- Stay active and hydrated

⊗ Vitamin B12

⊗ Folate

⊗ Iron

⊗ Vitamin C

✓ Eggs

✓ Leafy green vegetables

✓ Citrus fruits

✓ Legumes

Additional Epigenetic Biomarkers

HbA1c

Normal

080

64.2%

Function

HbA1c: Glycated hemoglobin; indicates average blood sugar levels over the past 2–3 months.

Recommendations

• Consume low-glycemic balanced meals

• Exercise regularly

• Pursue weight loss if applicable

• Reduce sugar intake

• Manage stress

• Consider medications if advised by your physician

• Consider supplements like ALA, berberine, or dihydroberberine

Alpha Lipoic Acid

Berberine

Dihydroberberine

Lean proteins

Healthy fats

Fiber-rich foods

Hematocrit

Normal

20100

92.8%

Function

Hematocrit: Percentage of red blood cells in blood; used to assess anemia and hydration.

Recommendations

• Correct nutrient deficiencies, especially folate, iron, B12, and vitamin C

• Treat illness

• Stay active and hydrated

Vitamin B12

Folate

Iron

Vitamin C

Eggs

Leafy green vegetables

Citrus fruits

Legumes

Blood Urea Nitrogen

Normal

080

35.5%

Function

Blood Urea Nitrogen (BUN): Measures urea levels in the blood, indicating kidney function and protein metabolism.

Recommendations

• Hydrate regularly

• If applicable, moderate protein intake

• Manage any kidney or liver disease

• Avoid nephrotoxic drugs

Alkaline Phosphatase

Normal

080

42.1%

Function

Alkaline Phosphatase: An enzyme found in the liver, bones, and other tissues; elevated levels indicate liver or bone issues.

Recommendations

• Correct Vitamin D deficiency

• Treat liver or bone disease if applicable

• Eat a diet low in saturated fats

• Exercise regularly

• Avoid alcohol and toxins

Vitamin D

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Additional Epigenetic Biomarkers

Insulin-like Growth Factor-Binding Protein 2

Normal

080

61.9%

Function

Insulin-Like Growth Factor-Binding Protein 2 (IGFBP2): Regulates insulin-like growth factor activity; linked to metabolism and aging.

Recommendations

Regularly practice calorie restriction under healthcare provider guidance

Engage in aerobic and resistance exercise to improve insulin sensitivity

Consume balanced meals

Manage stress

Reduce excessive protein intake

Improve sleep to support growth hormone function

Carboxypeptidase B2

Normal

080

15.1%

Function

Carboxypeptidase B2: CPB2 helps balance blood clotting and inflammation. When it's in a healthy range, it supports good circulation and efficient recovery.

Recommendations

Exercise regularly

Manage stress

Improve sleep

Avoid smoking

Eat more plan polyphenols, like berries, olive oil, green tea, and tumeric

Limit refined carbs and sugar

Quercetin

Magnesium

Serum Paraoxonase/arylesterase 1

Normal

20100

100%

Function

Serum Paraoxonase/Arylesterase 1 (PON1): An enzyme with antioxidant properties; linked to cardiovascular health.

Recommendations

Increase selenium intake with foods like Brazil nuts, seafood, and whole foods

Consider chokeberry extract and pomegranate

Minimize intake of trans fat

Chokeberry extract

Brazil nuts

Seafood

Pomegranate

Ribonuclease pancreatic

Normal

080

33.3%

Function

Ribonuclease Pancreatic: An enzyme that degrades RNA; linked to digestive and cellular processes.

Recommendations

Consider an anti-inflammatory diet

Regulate blood sugar levels

Manage stress

Maintain an active lifestyle

Omega-3 Fatty Acid

Anti-inflammatory diet

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Additional Epigenetic Biomarkers

Inter-alpha-trypsin Inhibitor Heavy Chain H3

Normal

080

35.0%

Function

Inter-Alpha-Trypsin Inhibitor Heavy Chain H3: A protein involved in inflammation and tissue repair.

Recommendations

Identify and control sources of inflammation, especially insulin resistance, gut microbiome imbalances, dietary factors, lipids, toxins, autoimmunity, and environmental exposures

Maintain an active lifestyle

Consider an anti-inflammatory diet

Anti-inflammatory diet

Ribitol

Normal

080

9.2%

Function

Ribitol: A sugar alcohol; involved in pentose phosphate and metabolic pathways.

Recommendations

Limit sugar and high glycemic food intake

Consider monitoring blood sugar

Lean protein

Low fat dairy

Vanillactic Acid

Normal

080

43.3%

Function

Vanillactate: A product of gut microbial metabolism of phenolic compounds; involved in energy pathways.

Recommendations

Manage blood pressure

Reduce sodium intake

Eat a heart-healthy diet

Reduce stress

Check heart function

Creatinine

Normal

080

61.1%

Function

Creatinine: A byproduct of muscle metabolism; used to evaluate kidney function.

Recommendations

Hydrate regularly

If applicable, reduce red meat intake

Control diabetes and hypertension

Avoid nephrotoxic drugs

Note that creatine can affect this level

Additional Epigenetic Biomarkers

Phenylacetylglutamine

Normal

080

58.0%

Function

Phenylacetylglutamine: A gut microbiome-derived metabolite of phenylalanine; linked to kidney function.

Recommendations

• Balance protein intake

• Limit foods high in phenylalanine (e.g, processed foods, dairy, meat, and some grains)

• Stay hydrated

• Monitor liver and kidney function

• Consider prebiotics or probiotics for gut health

Prebiotics

Probiotics

Uridine

Normal

20100

34.2%

Function

Uridine: A nucleoside involved in RNA synthesis and energy metabolism.

Recommendations

• Incorporate uridine-rich foods like tomatoes, broccoli, and mushrooms, which are synergistic with choline and DHA

Uridine

Choline

DHA

Tomatoes

Broccoli

Mushrooms

Eggs

Fatty fish

1-margaroyl-glycerophospholipid(GPE)(17:0)*

Normal

080

41.0%

Function

1-Margaroyl-Glycerophospholipid (GPE) (17:0): A glycerophospholipid; essential for cell membrane stability.

Recommendations

• If applicable, reduce saturated fat intake

• Increase omega-3 intake

• Consider anti-inflammatory agents like curcumin or green tea extract

Omega-3 Fatty Acid

Curcumin

Green tea extract

Green tea

4-Methoxyphenol Sulfate

Normal

080

39.2%

Function

4-Methoxyphenol Sulfate: A sulfated metabolite of phenolic compounds; reflects detoxification processes.

Recommendations

• Avoid woodsmoke exposure and dark-aged beer

• Avoid synthetic fragrances

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Additional Epigenetic Biomarkers

N-acetyl-isoputresnine Normal 0 80 28.6%	Function N-Acetyl-Isoputresnine: A metabolite of polyamine biosynthesis; associated with cellular growth and repair. Recommendations • Consider NAC supplementation NAC
Gluconate Normal 0 80 67.5%	Function Gluconate: A metabolite of glucose; involved in carbohydrate metabolism and energy production. Recommendations • Cut added sugar, added fat, and processed foods • Exercise regularly • Consider ALA, berberine, or dihydroberberine • Limit wine Alpha Lipoic Acid Berberine Dihydroberberine
Carotene Diol Normal 20 100 58.2%	Function Carotene Diol: A carotenoid metabolite; an antioxidant linked to vitamin A activity. Important for eye and brain health. Recommendations • Consume more carotenoid-rich foods like carrots, kale, and spinach Carrots Kale Spinach
4-hydroxyphenylacetylglutamine Normal 0 80 35.2%	Function 4-Hydroxyphenylacetylglutamine: A gut microbial metabolite of tyrosine; linked to gut health and kidney function. Recommendations • Manage blood pressure • If applicable, reduce protein intake • Increase fiber and fermented foods • Consider prebiotics or probiotics • Stay hydrated • Protect kidney health Prebiotics Probiotics Kefir Yogurt Sauerkraut Fiber-rich foods

Additional Epigenetic Biomarkers

Androsterone Sulfate

Normal

20

100

65.5%

Function

Androsterone Sulfate: A steroid hormone metabolite downstream of testosterone; reflects androgen activity and metabolism.

Recommendations

- Correct any nutrient deficiencies, especially zinc, magnesium, vitamin D, and omega-3
- Optimize sleep
- Manage stress
- Consider hormone therapy if needed
- Maintain an active lifestyle

Zinc

Magnesium

Vitamin D

Omega-3

DHEA

3-Ureidopropionate

Normal

0

80

1.7%

Function

3-Ureidopropionate: A product of pyrimidine metabolism; involved in nitrogen recycling.

Recommendations

- If applicable, reduce excessive protein intake, especially those rich in pyrimidines like organ meats
- Stay hydrated
- Monitor kidney and liver health

Bone Morphogenetic protein 1

Normal

20

100

95.3%

Function

Bone Morphogenetic Protein 1 (BMP1): A protein involved in bone formation and extracellular matrix remodeling.

Recommendations

- Consume calcium-rich and vitamin D-rich foods
- Consider calcium or vitamin D supplementation
- Increase weight-bearing exercise
- Ensure adequate protein intake

Calcium

Vitamin D

Green leafy vegetables

Nuts

Dairy

Fatty fish

Versican core protein

Normal

0

80

19.6%

Function

Versican Core Protein: A proteoglycan in the extracellular matrix; involved in tissue remodeling and inflammation.

Recommendations

- Avoid being underweight
- Increase moderate-intensity exercise

Additional Epigenetic Biomarkers

Histone H2B type 1-K

Normal

▼

0

80

5.9%

Function

Histone H2B Type 1-K: A histone protein involved in chromatin structure and epigenetic regulation.

Recommendations

- Limit UV exposure
- Increase antioxidant intake
- Avoid smoke exposure
- Manage stress
- Maintain an active lifestyle

1-Stearyl-2-adrenoyl-GPC (18:0/22:4)*

Normal

▼

20

100

33.6%

Function

A glycerophosphocholine (GPC) lipid, this compound supports membrane integrity, brain function, and lipid metabolism, essential for longevity. Low levels may impair neuroprotection, increase oxidative stress, and accelerate age-related decline.

Recommendations

- If applicable, correct omega-3 deficiency
- Increase antioxidants
- Limit alcohol

✎ Omega-3 Fatty Acid

✓ Flaxseeds

✓ Walnuts

✓ Fatty fish

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